

Chicago Crime & Weather Analysis

Exploratory Data Analysis Report

Objective

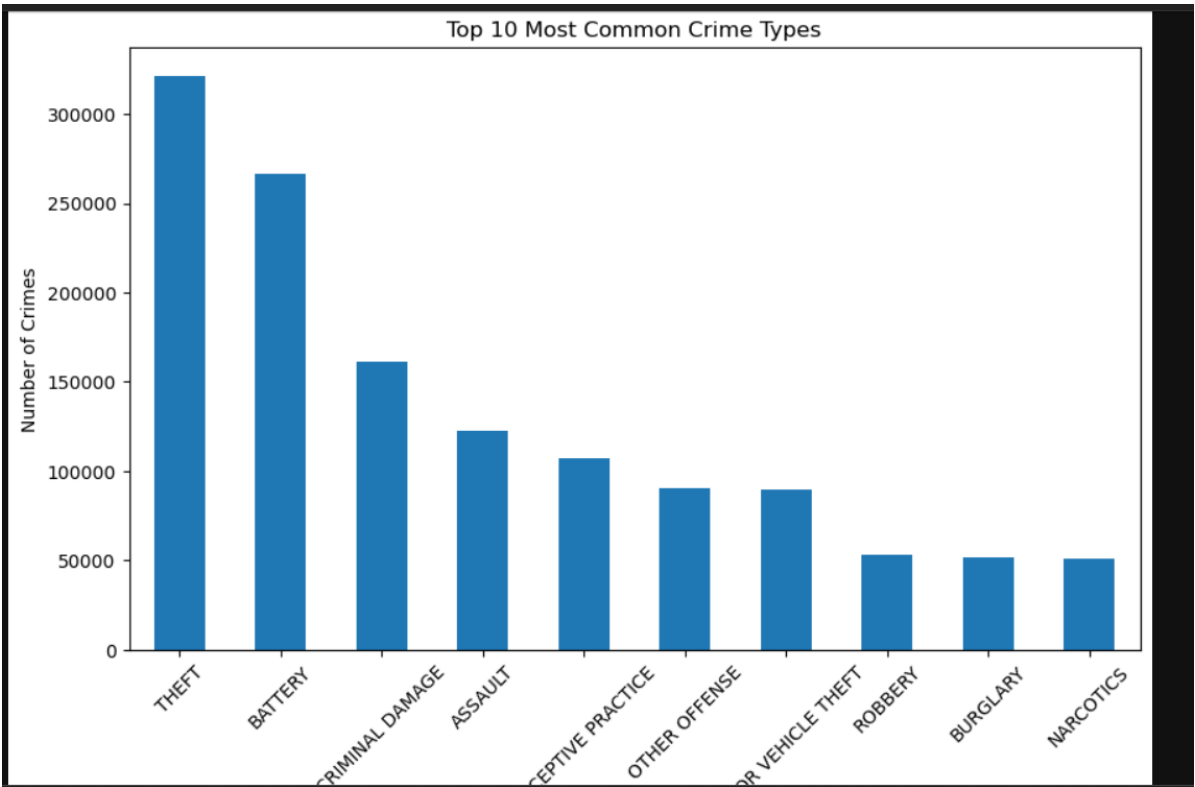
Analyze Chicago crime, weather, and community datasets to identify crime trends, seasonal patterns, weather relationships, arrest outcomes, and community-level insights using PostgreSQL and Python.

Tools Used

PostgreSQL, Python, Pandas, NumPy, Matplotlib

Top 10 Most Common Crime Types

Purpose: Identify the most frequently reported crime categories.



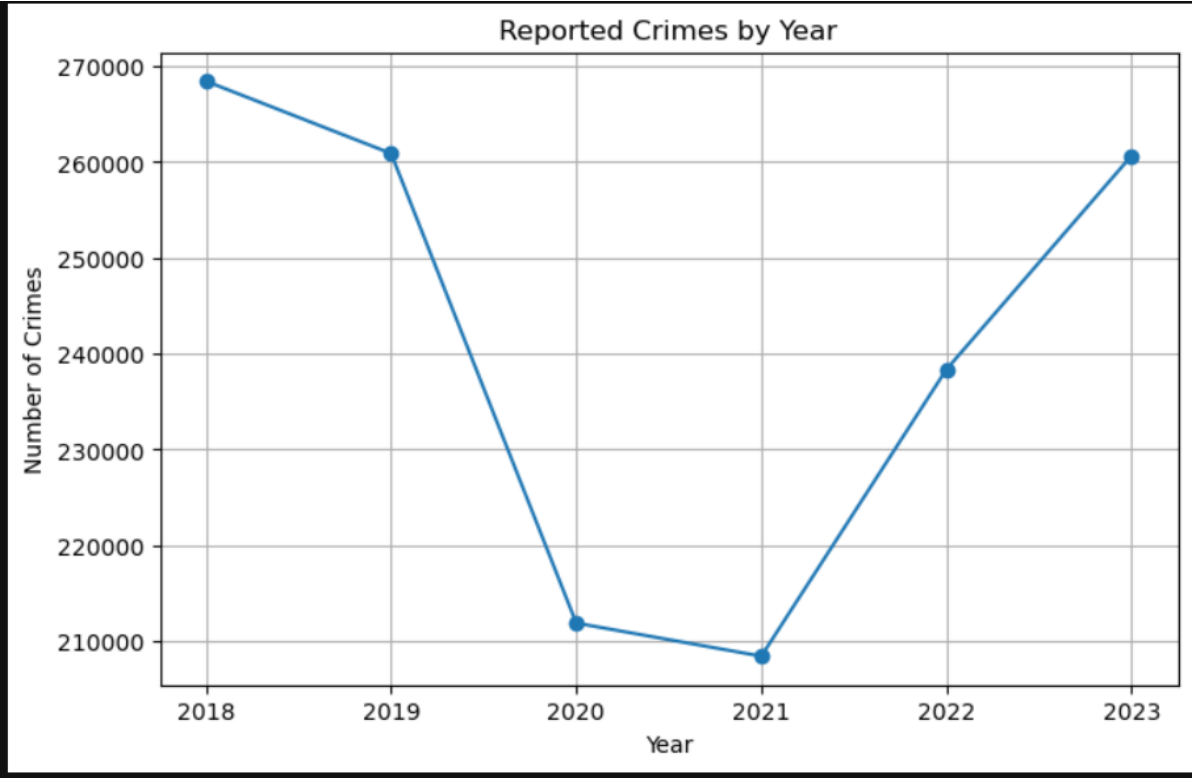
Key Findings

- Theft is the most frequently reported crime type.
- Battery ranks second, followed by Criminal Damage and Assault.
- Property-related crimes account for a significant share of reported incidents.

Business Insight: Knowing the dominant crime categories helps prioritize policing strategies and prevention programs.

Reported Crimes by Year

Purpose: Analyze long-term crime trends.



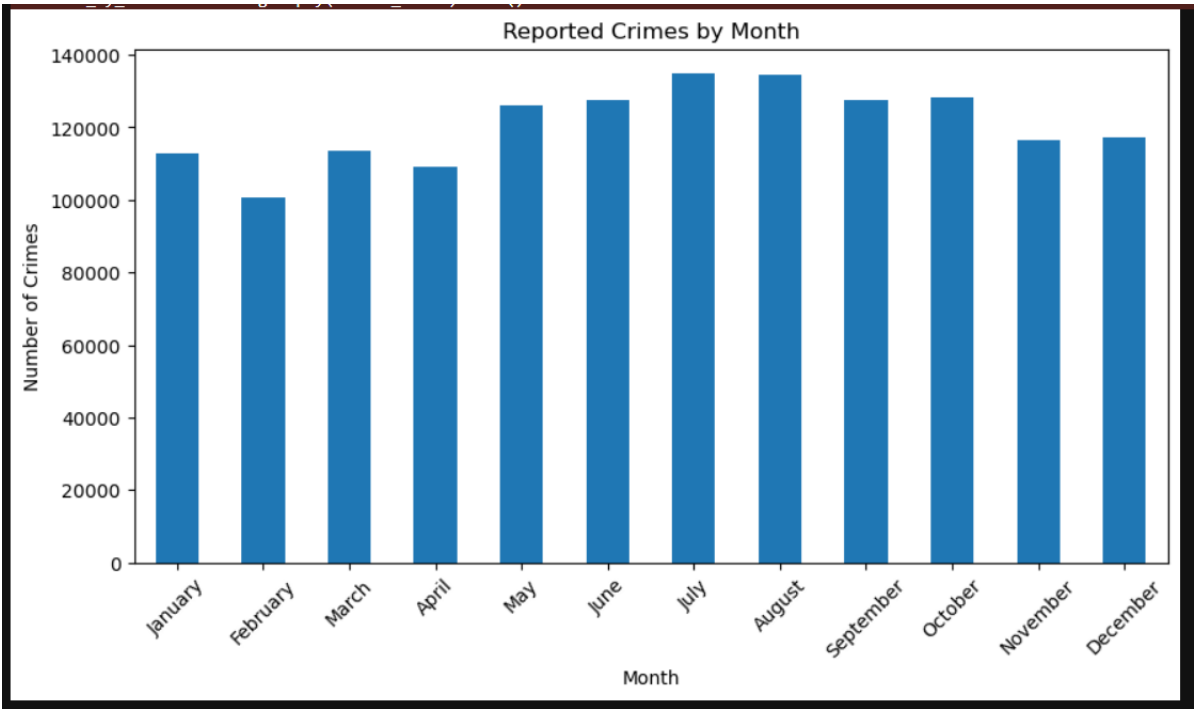
Key Findings

- Crime declined during 2020–2021.
- Reported incidents increased again in 2022–2023.
- The trend suggests recovery toward pre-2020 levels.

Business Insight: Yearly trend analysis helps evaluate changes in crime patterns over time.

Reported Crimes by Month

Purpose: Examine seasonal crime patterns.



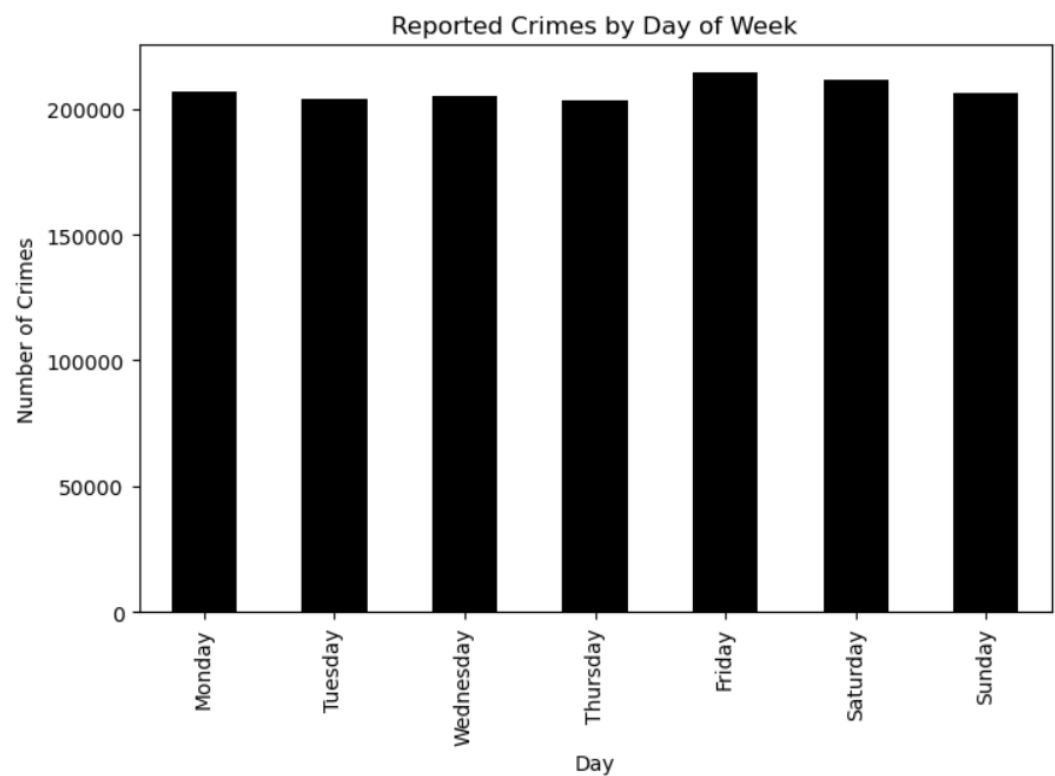
Key Findings

- Crime generally increases from May onward.
- July and August record the highest crime counts.
- Winter months show comparatively lower activity.

Business Insight: Seasonal trends can support resource planning during high-crime periods.

Reported Crimes by Day of Week

Purpose: Compare crime frequency across weekdays.



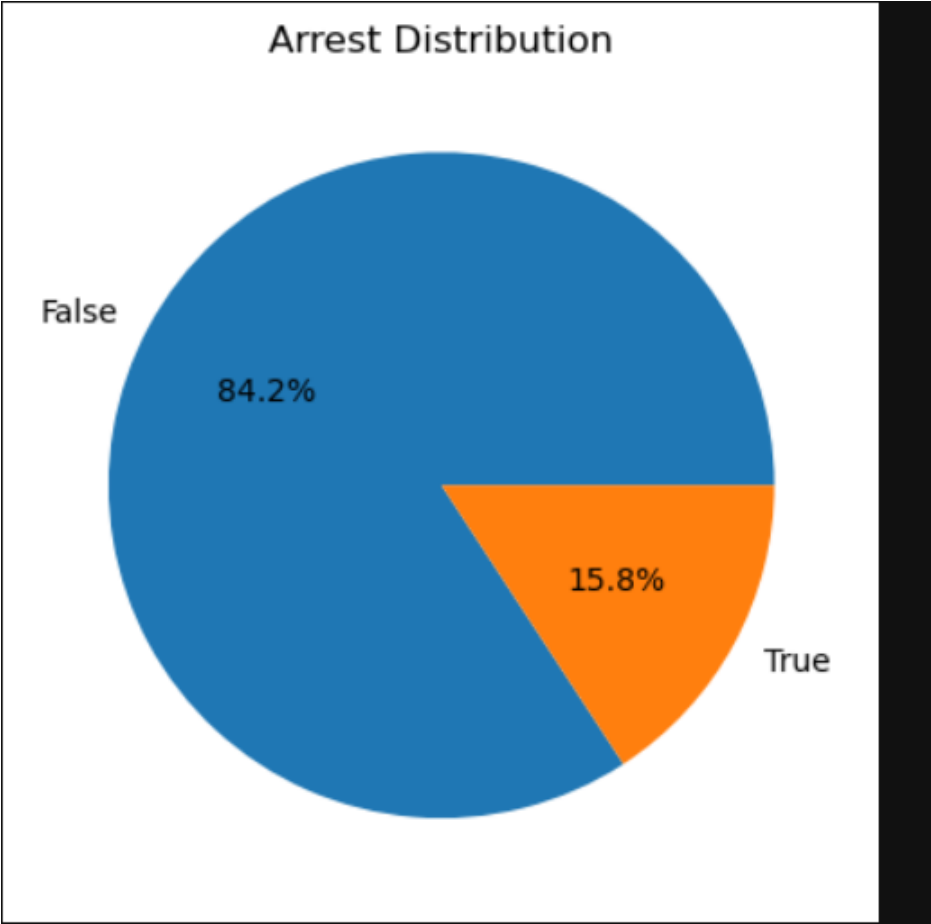
Key Findings

- Crime levels remain relatively consistent throughout the week.
- Friday and Saturday show slightly higher crime activity.

Business Insight: Weekend resource allocation may improve response efficiency.

Arrest Distribution

Purpose: Measure arrest outcomes.



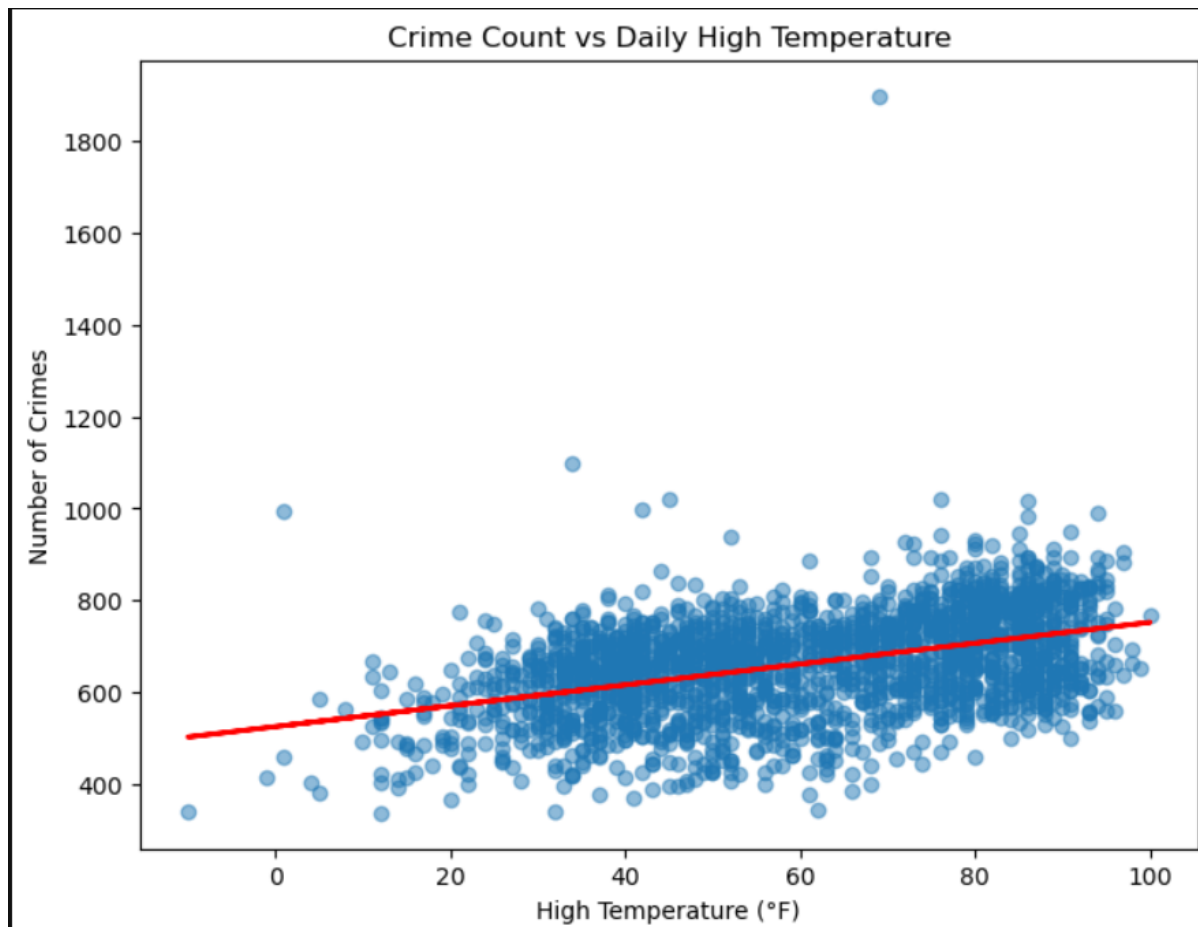
Key Findings

- 84.2% of reported crimes did not result in an arrest.
- 15.8% resulted in an arrest.

Business Insight: The large difference highlights opportunities to improve case resolution.

Crime Count vs Daily High Temperature

Purpose: Study the relationship between weather and crime.



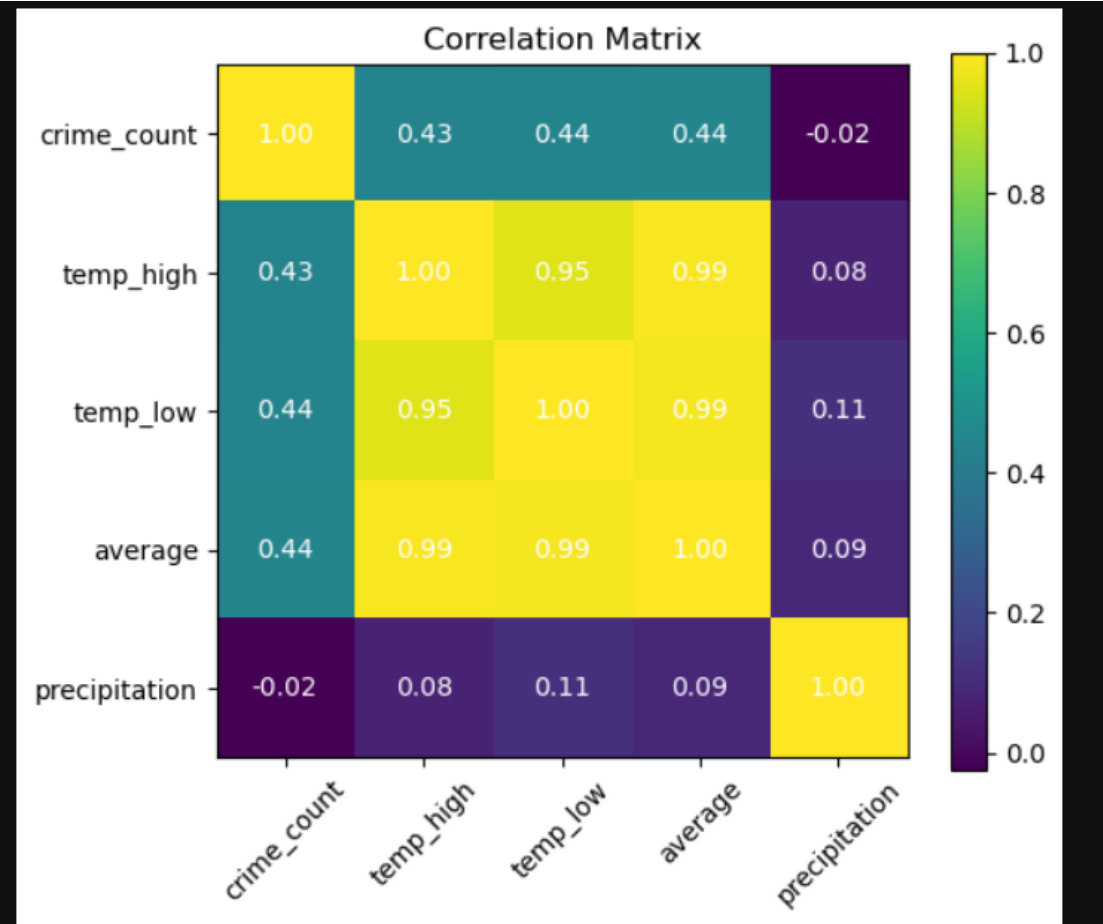
Key Findings

- A weak positive relationship exists between temperature and crime count.
- Warmer days generally experience slightly more crimes.
- A few outliers are present.

Business Insight: Temperature alone is not sufficient to explain crime patterns.

Correlation Matrix

Purpose: Measure linear relationships among variables.



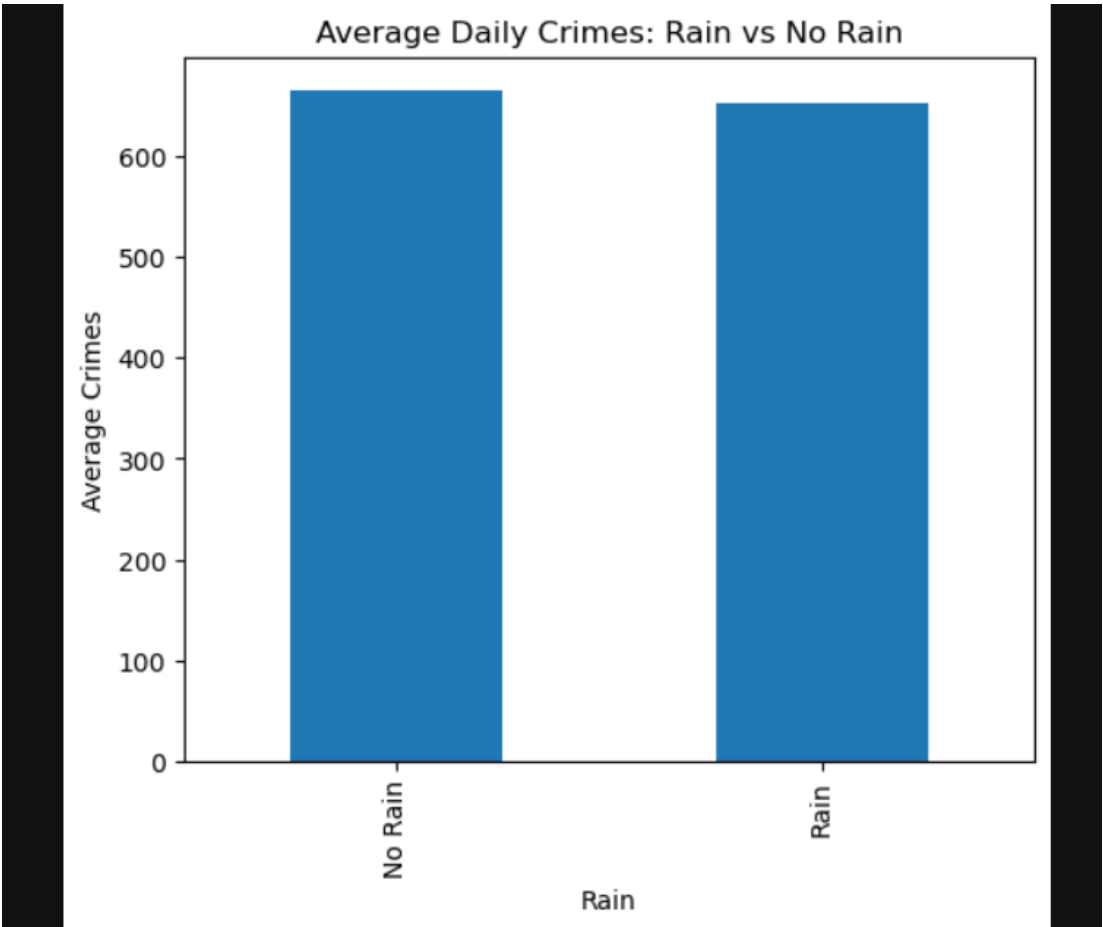
Key Findings

- Temperature variables are highly correlated.
- Crime count has a moderate positive correlation with temperature (~0.43–0.44).
- Precipitation shows almost no linear relationship with crime count.

Business Insight: Weather contributes to crime patterns but is not the only influencing factor.

Average Daily Crimes: Rain vs No Rain

Purpose: Compare crime levels on rainy and dry days.



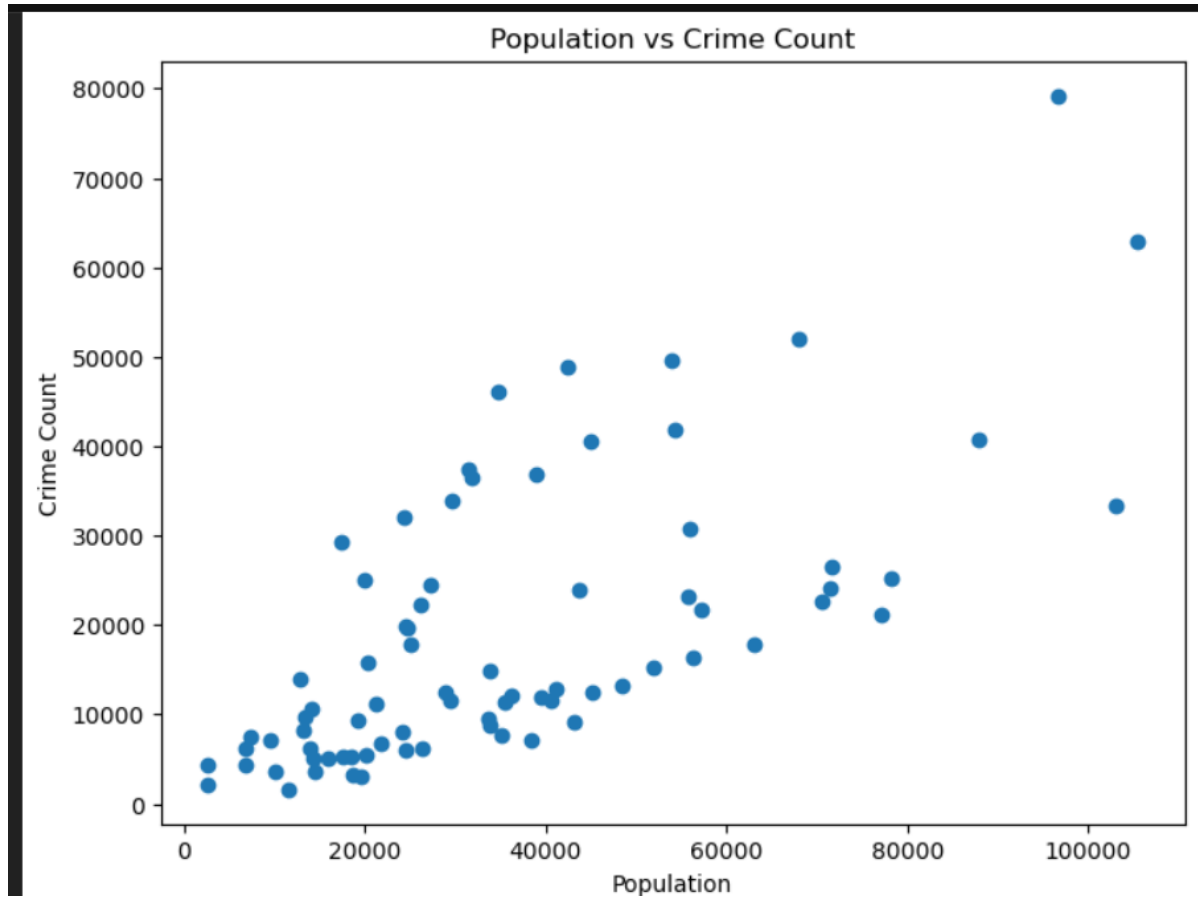
Key Findings

- Average daily crime counts are very similar.
- Rainfall alone has minimal influence on daily crime volume.

Business Insight: Other variables beyond rainfall are likely more important.

Population vs Crime Count

Purpose: Compare community population with reported crimes.



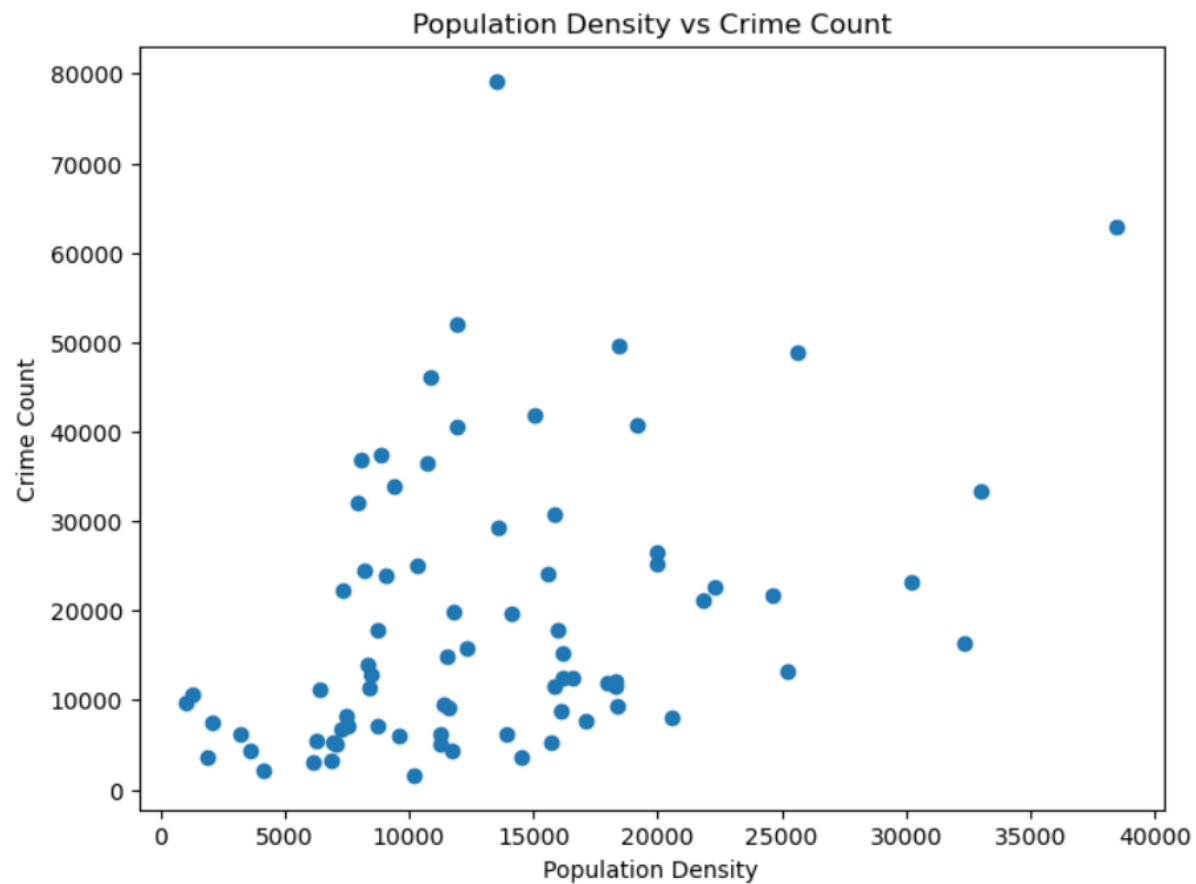
Key Findings

- Communities with larger populations generally report more crimes.
- Population alone does not fully explain crime differences.

Business Insight: Socioeconomic and geographic factors should also be considered.

Population Density vs Crime Count

Purpose: Assess whether denser communities experience more crime.



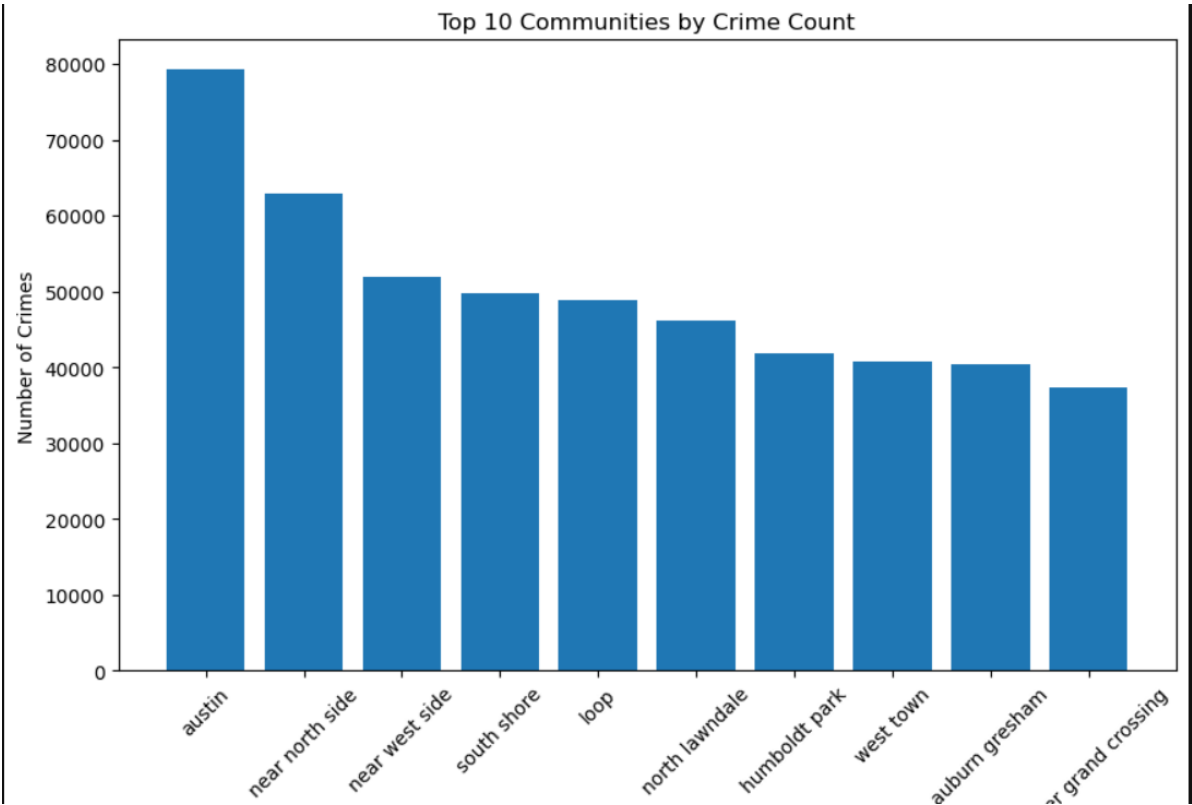
Key Findings

- No strong relationship is visible.
- High density does not always correspond to higher crime counts.

Business Insight: Population density is not a standalone predictor of crime.

Top 10 Communities by Crime Count

Purpose: Identify communities with the highest crime volume.



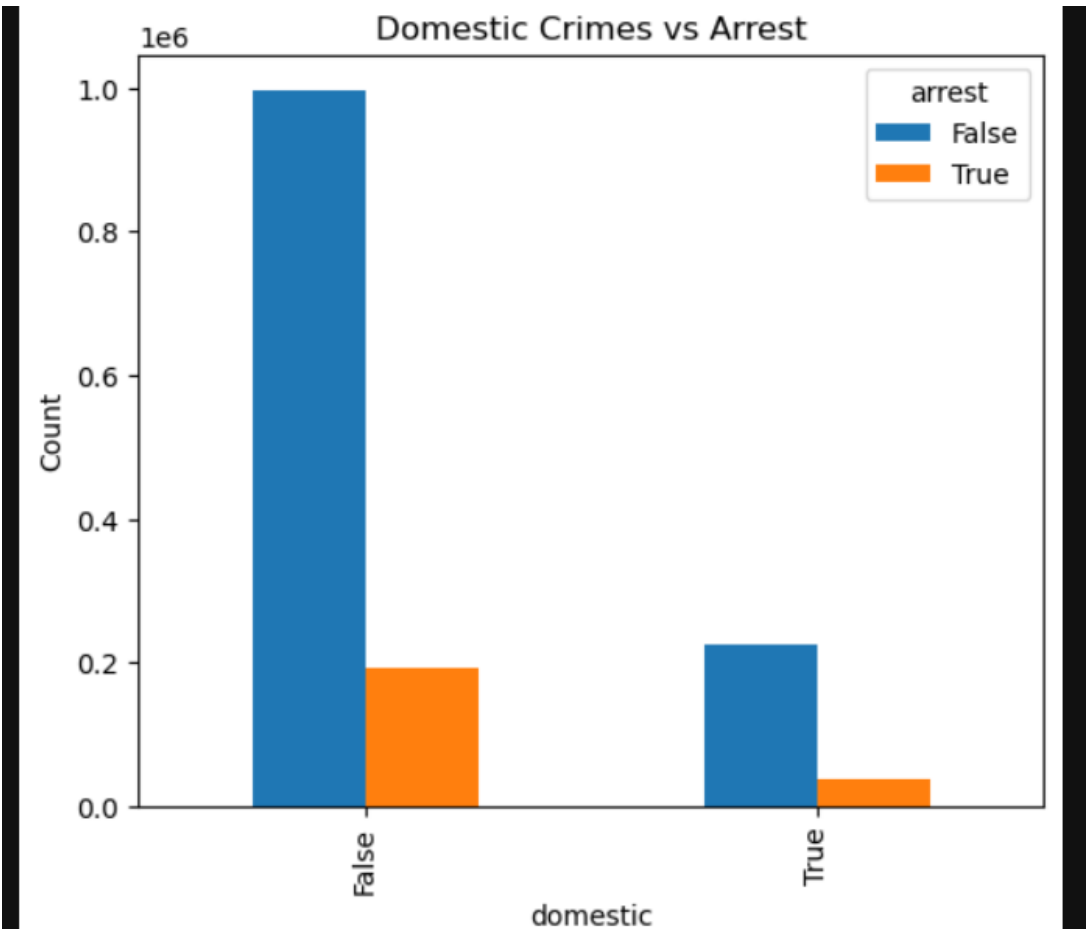
Key Findings

- Austin recorded the highest crime count.
- Near North Side and Near West Side follow closely.

Business Insight: High-crime communities may require targeted interventions.

Domestic Crimes vs Arrest

Purpose: Compare arrest outcomes for domestic and non-domestic crimes.



Key Findings

- Non-arrest cases exceed arrest cases in both categories.
- Non-domestic crimes are more common than domestic crimes.

Business Insight: Understanding arrest outcomes supports evaluation of enforcement effectiveness.

Overall Conclusion

The analysis indicates that theft remains the dominant crime category, crime activity is generally higher during warmer months, and temperature has a moderate positive relationship with daily crime counts. Rainfall has minimal influence on crime levels, while community characteristics such as population contribute to crime variation. These findings can support evidence-based resource allocation and public safety planning.